

Mouse auditory experiment

Simple setup draft

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Goal. Play known tones to a mouse and record the auditory cortical response, with a reliable timestamp for each sound. This is a proposed mouse adaptation of SONIC; awake versus anesthetized state remains undecided.

1. Define the tones

Control	SONIC reference	Our mouse draft
Frequency	Explored 1.4–32 kHz	Select a frequency set
Tone duration	10 ms in benchmark sessions	Adjustable duration
Sound level	Approximately 80 dB SPL	Choose a measured target
Gap between tones	0 ms within each block	Adjustable silent gap
Start/end ramps	2 ms at each end	Define shape and duration

Paper values come from awake sheep. Mouse settings remain open; the paper does not specify the ramp shape. Save the tone sequence in advance, with a trial number and settings for every tone.

2. Set up the sound path

Saved tones → DAC/audio output → amplifier → speaker → mouse

Evaluate the **TDT MF1 speaker and SA1 amplifier** used in SONIC, starting with available lab equipment. Check the full chain’s frequency coverage. Keep speaker placement fixed and use a suitable calibrated microphone at the ear reference position to measure each selected frequency’s sound level.

3. Align sound and neural recordings

Use hardware-timed playback to generate each tone and an electrical timing marker on the same schedule. Capture the marker with the neural recording. Record the microphone signal with a verified clock relationship to the neural data, then measure the delay from marker to actual sound arrival.

Save trial number, frequency, duration, level, gap, ramp settings, marker time, and measured acoustic onset. If behavior video is added, align camera frame timing to this same timeline. Exact connections depend on the recorder and camera models.

4. Start in three steps

1. Identify the available speaker, microphone, audio output, and neural recorder.
2. On the bench, check delivered tones, levels, timing, and electrical pickup in the recording path.
3. With the team’s mouse protocol, characterize responses to separated tones before choosing a faster sequence and response window.

Still open: mouse strain/age and state, final tone settings, recorder, and any behavior task. A simple enclosure may help repeatable positioning; current team direction does not require sound isolation.

Sources: Perkins et al., SONIC, bioRxiv v1 (2025), Methods 6.1–6.2, p.11, and 6.6, p.13. Hardware and timing guidance: TDT MF1 specifications; TDT I/O delays. Microphone onset checks and the separated-tone progression are proposed Rice additions.